



**Southeast University**

Department of Computer Science and Engineering

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## **PROJECT REPORT**

### **Interactive 3D Classroom**

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**Course Title:** Computer Graphics & Animation Lab

**Course Code:** CSE444.2

#### **Submitted By**

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## 1. Introduction

Computer graphics is an important area of computer science used to create, manipulate, and display visual objects and virtual environments. Three-dimensional graphics make it possible to represent real-world environments using geometric models, transformations, camera systems, lighting, colors, textures, and animation.

This project presents an **Interactive 3D Classroom** developed using C++ and OpenGL. The final classroom contains desks, chairs, computer workstations, walls, a tiled floor, ceiling, windows, a classroom door, projector, projection screen, wall switches, lighting systems, and other classroom objects.

The project is designed not only as a static 3D scene but also as an interactive environment. The user can move around the classroom, change the viewing direction, interact with the door, operate classroom lights, control computers, and use the projector.

The final implementation also includes directional sunlight, real-time shadows, textured materials, projector effects, computer states, audio feedback, and multiple interactive controls.

The main academic requirements of the project are:

- 3D transformation.
- Viewing transformation.
- At least one moving object.
- At least two different kinds of light.
- Different colors for different objects.

The project combines these fundamental computer graphics concepts with additional interactive features to create a more complete and realistic classroom environment.

## 2. Objectives

The main objectives of the Interactive 3D Classroom project are:

- To create an interactive three-dimensional classroom environment.
- To implement translation, rotation, and scaling transformations.
- To implement viewing transformation using a movable camera.
- To demonstrate object animation using an interactive classroom door.
- To implement multiple lighting techniques including point lights, spotlight, and directional sunlight.
- To implement visible shadow effects for improved depth perception.
- To apply different colors, textures, and material properties to classroom objects.
- To implement interactive classroom computers.
- To implement realistic wall switches for classroom lighting control.
- To implement an interactive projector and projection screen.
- To provide keyboard and mouse based user interaction.

- To gain practical experience with C++, OpenGL, GLM, GLSL shaders, 3D models, lighting, animation, and real-time rendering.

### 3. Tools & Technologies

The project was developed using a combination of C++ programming, OpenGL libraries, mathematical libraries, shaders, model loading, audio processing, and build tools.

| Technology    | Use in the Project                                                                                                 |
|---------------|--------------------------------------------------------------------------------------------------------------------|
| C++           | Main programming language used to implement the application, classroom objects, interactions, and rendering logic. |
| OpenGL        | Used to render the interactive three-dimensional classroom.                                                        |
| GLFW          | Used for window creation and keyboard and mouse input.                                                             |
| GLAD          | Loads the required OpenGL functions.                                                                               |
| GLM           | Provides vectors, matrices, transformations, and camera calculations.                                              |
| GLSL          | Used to implement vertex and fragment shaders, lighting, materials, and rendering effects.                         |
| Assimp        | Used for loading external 3D model assets and their associated model information.                                  |
| miniaudio     | Used to provide audio and interactive sound feedback.                                                              |
| CMake         | Used to configure the project, manage dependencies, and generate the Visual Studio solution.                       |
| Visual Studio | Used to write, compile, debug, build, and run the C++ project.                                                     |

## 4. Theoretical Background

### 4.1 3D Transformation

3D transformations control the position, orientation, and size of objects in a virtual environment. Three fundamental transformations are used in the project: **translation**, **rotation**, and **scaling**.

Translation moves an object from one location to another. Rotation changes the orientation of an object. Scaling changes its dimensions.

The project uses GLM functions such as:

- `glm::translate()`
- `glm::rotate()`
- `glm::scale()`

These transformations are combined inside model matrices to position classroom objects correctly.

### 4.2 Viewing Transformation

Viewing transformation determines how the three-dimensional environment is observed.

The project uses a movable camera system. The camera generates a view matrix using `glm::lookAt()`.

The camera position, front direction, and up vector determine the user's view of the classroom.

Keyboard input changes the camera position while mouse movement changes the viewing direction.

### 4.3 Lighting, Shadows, and Colors

Lighting improves visibility, depth perception, and realism in a three-dimensional environment.

The classroom uses multiple lighting techniques.

**Point lights** provide local classroom illumination.

A **spotlight** is used to create the directional projector lighting effect.

A **directional light** represents sunlight entering through the classroom windows.

The directional sunlight is combined with shadow rendering so that classroom objects can produce visible shadows. This improves the visual depth of the classroom.

Different RGB colors, textures, material properties, and lighting values are used for the floor, walls, ceiling, desks, computers, projector, switches, and other classroom objects.

#### 4.4 Texture and Material Rendering

Textures are used to improve the visual appearance of surfaces. The classroom uses textured material information for selected objects such as the floor.

Material properties control how an object reacts to light. Different surface properties help distinguish metal, plastic, painted surfaces, screens, and other classroom components.

#### 4.5 Animation

Animation is produced by changing object properties over time.

The classroom door demonstrates rotation-based animation. Its rotation angle changes during the opening and closing sequence.

The project also contains state-based visual transitions for interactive classroom devices.

### 5. System Design

The system is divided into several components responsible for application control, classroom rendering, camera movement, transformations, lighting, shadows, animation, interactive objects, audio, shaders, and final rendering.

The modular structure allows different parts of the classroom to be controlled independently while still being combined into one interactive environment.

#### 5.1 System Architecture

The overall architecture of the system is shown below.

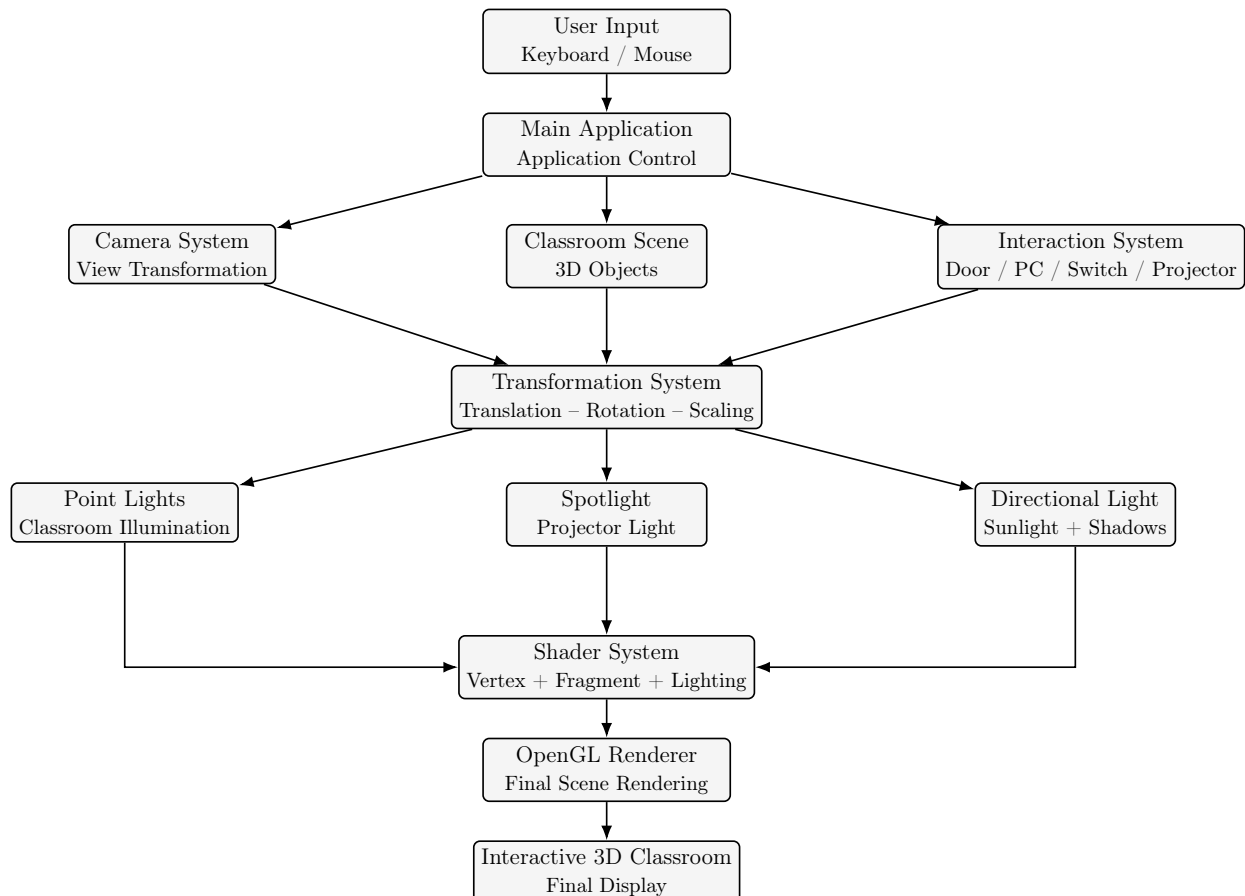


Figure 1: Architecture of the Interactive 3D Classroom

## 6. Interactive Features

The final classroom contains several interactive elements. These features make the application more than a static three-dimensional scene.

- The user can move around the classroom using the camera.
- Mouse movement changes the viewing direction.
- The classroom door can be opened and closed.
- Classroom computers support interactive power states.
- Wall switches control classroom lighting.
- Two realistic wall switches are included in the classroom.
- The projector can be turned on and off.
- The projector provides a visible projection effect.
- A presentation mode displays a **WELCOME** message on the projection screen.
- Directional sunlight enters through the windows.
- Shadows improve the depth and appearance of the classroom.
- Interactive actions can provide audio feedback.

### 6.1 User Controls

The application supports keyboard and mouse based interaction.

| Control    | Function                                |
|------------|-----------------------------------------|
| W, A, S, D | Move the camera through the classroom.  |
| Mouse      | Change the camera viewing direction.    |
| Shift      | Increase camera movement speed.         |
| E          | Interact with nearby classroom objects. |
| L          | Toggle classroom lighting.              |
| F1 / H     | Enable or disable shadow rendering.     |
| F4         | Power on classroom computers.           |
| F5         | Shut down classroom computers.          |
| F6         | Reset computers to the off state.       |
| F7 / O     | Toggle projector power.                 |
| F8 / P     | Change projector or presentation mode.  |
| Esc        | Release or capture the mouse cursor.    |

## 7. Animations Used

### 7.1 Door Animation

The main moving object required by the project is the **classroom door**.

The door animation is produced by continuously changing its rotation angle.

The function:

```
Door::Update(deltaTime)
```

updates the animation over time.

The variable:

```
m_currentAngle
```

stores the current rotation angle.

The door rotates around the vertical Y-axis. As the angle increases or decreases, the door appears to open or close.

This demonstrates how rotation transformation can be combined with frame-based updates to create movement in a three-dimensional environment.

## 7.2 Additional Interactive Transitions

Besides the classroom door, the project contains state-based visual transitions for interactive classroom devices.

Computers can change between power states, while the projector can change between inactive and active states.

Lighting conditions also change when the classroom switches are operated.

These additional transitions improve the interactive appearance of the environment while the classroom door remains the primary moving object required by the project.

## 8. Implementation

This section describes the main implementation techniques used in the Interactive 3D Classroom. Each subsection briefly explains the concept and shows the relevant implementation approach used in the project.

### 8.1 Transformation Implementation

**Definition:** A transformation changes the position, orientation, or size of an object in a three-dimensional environment. The three basic transformations are translation, rotation, and scaling.

In the project, GLM matrices are used to transform classroom objects.

```
1 glm::mat4 model =
2     glm::translate(
3         glm::mat4(1.0f),
4         position
5     );
6
7 model = glm::rotate(
8     model,
9     rotationY,
10    glm::vec3(0.0f, 1.0f, 0.0f)
11 );
12
13 model = glm::scale(
14     model,
15     scale
16 );
```

**Explanation:** `glm::translate()` moves an object to its required position. `glm::rotate()` changes its orientation, and `glm::scale()` changes its size.

These transformations are combined into a model matrix. In the classroom, transformations are used to position and size desks, chairs, computers, the projector, switches, and other objects. Rotation is also important for the animated classroom door.

### 8.2 Viewing Implementation

**Definition:** Viewing transformation determines how the three-dimensional scene is observed through the virtual camera. The view matrix represents the position and orientation of the camera.

The camera system generates the view matrix using `glm::lookAt()`.

```
1 glm::mat4 Camera::GetViewMatrix() const
2 {
3     return glm::lookAt(
4         Position,
```

```

5         Position + Front,
6         Up
7     );
8 }

```

The generated view matrix is then supplied to the scene shader.

```

1 glm::mat4 view = camera.GetViewMatrix();
2 sceneShader.setMat4("view", view);

```

**Explanation:** `Position` represents the current camera location, `Front` represents the viewing direction, and `Up` represents the camera's upward direction.

Keyboard input changes the camera position, while mouse movement changes the viewing direction. Therefore, the user can explore the classroom from different positions and angles.

### 8.3 Lighting Implementation

**Definition:** Lighting determines how visible surfaces are illuminated in a 3D scene. Using different types of light improves depth perception and makes the classroom environment more realistic.

The project uses multiple lighting techniques, including point lighting, projector spotlight, and directional sunlight.

A simplified representation of the lighting data is:

```

1 struct PointLight
2 {
3     glm::vec3 position;
4     glm::vec3 color;
5     float intensity;
6 };
7
8 struct SpotLight
9 {
10    glm::vec3 position;
11    glm::vec3 direction;
12    glm::vec3 color;
13 };

```

Lighting information is supplied to the shader before the classroom objects are rendered.

```

1 sceneShader.setVec3(
2     "viewPos",
3     camera.Position
4 );
5
6 // Light information is then passed
7 // to the scene shader for rendering.

```

**Explanation:** Point lights provide local illumination inside the classroom. The projector uses directional spotlight-like illumination, while directional lighting represents sunlight entering through the windows.

Therefore, the final project uses more than the minimum requirement of two different lighting techniques.

### 8.4 Shadow Implementation

**Definition:** A shadow is produced when an object blocks light from reaching another surface. Shadows provide important visual information about the relative positions of objects and improve depth perception.

The classroom uses directional sunlight together with a shadow rendering stage. A light-space transformation is used to represent the scene from the direction of the light.

A simplified form of the required transformation is:

```

1 glm::mat4 lightSpaceMatrix =

```

```
2 | lightProjection * lightView;
```

The light-space matrix is then provided to the shader.

```
1 | sceneShader.setMat4(
2 |     "lightSpaceMatrix",
3 |     lightSpaceMatrix
4 | );
```

During rendering, depth information from the light's point of view can be compared with the current fragment position.

**Explanation:** If a surface is blocked from the directional light by another object, it appears darker. This allows desks, chairs, computers, and other classroom objects to produce visible shadows when illuminated by sunlight.

## 8.5 Computer Workstation Implementation

**Definition:** A computer workstation is a group of 3D objects representing a complete classroom computer setup. Each workstation contains a monitor, keyboard, mouse, and computer tower.

The computer components are positioned using model transformations.

A simplified placement pattern is:

```
1 | glm::mat4 model(1.0f);
2 |
3 | model = glm::translate(
4 |     model,
5 |     workstationPosition
6 | );
7 |
8 | model = glm::scale(
9 |     model,
10 |    workstationScale
11 | );
```

The computer tower uses a solid dark material so that it remains clearly visible.

```
1 | glm::vec3 towerColor(
2 |     0.05f,
3 |     0.05f,
4 |     0.06f
5 | );
```

**Explanation:** The computer tower is placed on top of the desk beside the monitor instead of underneath the table. The teacher and student workstations follow the same general computer arrangement.

The solid dark material prevents the tower from appearing visually transparent and provides a more realistic desktop computer setup.

## 8.6 Classroom Table Implementation

**Definition:** The classroom tables are 3D objects used to support the computers and other workstation components. Their position and dimensions are controlled using translation and scaling transformations.

A table component can be positioned using:

```
1 | glm::mat4 tableModel(1.0f);
2 |
3 | tableModel = glm::translate(
4 |     tableModel,
5 |     tablePosition
6 | );
7 |
8 | tableModel = glm::scale(
9 |     tableModel,
10 |    tableScale
```

11 );

**Explanation:** The teacher table and student tables use a consistent classroom design. The teacher workstation was adjusted so that its table follows the same general appearance as the other classroom tables.

The computer equipment is positioned above the table surface to avoid unwanted intersections between the keyboard, mouse, monitor, and table geometry.

Unnecessary lower support geometry was removed from the teacher table to provide a cleaner layout.

## 8.7 Floor Material Implementation

**Definition:** A material determines the visual surface properties of a 3D object. Color and texture information can be used together to produce a more realistic surface.

The classroom floor uses a gray tiled appearance instead of a completely black surface.

A representative gray floor color can be expressed as:

```
1 glm::vec3 floorColor(
2     0.30f,
3     0.31f,
4     0.33f
5 );
```

The floor also uses texture information to provide visible surface detail.

```
1 // Floor material textures are loaded
2 // and used by the rendering system.
3 //
4 // Base texture
5 // Normal texture
6 // Roughness texture
```

**Explanation:** The gray tiled floor improves contrast between the floor and the classroom furniture. It also makes the scene brighter and more visually balanced than a completely black floor.

Normal and roughness information can provide additional surface detail when processed by the material and shader system.

## 8.8 Light Switch Implementation

**Definition:** A light switch is an interactive classroom object used to control the lighting state. The project contains two wall-mounted switches with a more realistic visual design.

The switch can store an ON/OFF state:

```
1 bool lightOn = true;
2
3 if (switchPressed)
4 {
5     lightOn = !lightOn;
6 }
```

The appearance of the rocker can also change according to the current state.

```
1 float rockerAngle =
2     lightOn ? -5.0f : 5.0f;
```

**Explanation:** Each switch is visually represented using a light-colored wall plate, a raised central rocker, and side details. This makes the switch look more like a physical classroom light switch instead of a simple flat rectangle.

The state of the switch is connected with the classroom lighting system, allowing the user to interact with the lights.

## 8.9 Projector System Implementation

**Definition:** The projector is an interactive classroom device that produces a directional projection effect and displays presentation content on the front screen.

The projector maintains a power state.

```
1 if (projectorOn)
2 {
3     // Render projector light.
4     // Render presentation content.
5 }
```

When presentation mode is active, the projection screen displays the WELCOME content.

A simplified state check is:

```
1 if (projectorOn && presentationMode)
2 {
3     RenderWelcome();
4 }
```

The projector direction can also be used by the lighting system.

```
1 glm::vec3 projectorDirection =
2     glm::normalize(
3         screenPosition -
4         projectorPosition
5     );
```

**Explanation:** When the projector is turned on, its directional light creates the projection effect toward the front screen.

In presentation mode, the screen displays the word **WELCOME**. This gives the user clear visual feedback that the projector is active and makes the classroom environment more interactive.

## 9. Results & Screenshots

The completed application successfully renders an interactive 3D classroom environment.

The final result demonstrates the main project requirements, including 3D transformations, viewing transformation, object animation, multiple lighting techniques, and different colors. Additional features such as directional sunlight, shadows, computer workstations, classroom switches, and projector interaction are also demonstrated in the final environment.

### 9.1 Final Classroom Environment

Figure 2 shows the overall final classroom environment. Different classroom objects are positioned using translation, rotation, and scaling transformations.

Different colors and materials are also used to visually distinguish the floor, walls, furniture, computers, and other classroom objects.



Figure 2: Overall View of the Final Interactive 3D Classroom

## 9.2 Door Animation Result

The classroom door is used as the main moving object of the project. The movement is created by changing the rotation angle of the door around its vertical axis.

The door can smoothly move between its closed and open positions. A video demonstration of the complete door opening and closing animation is provided below.

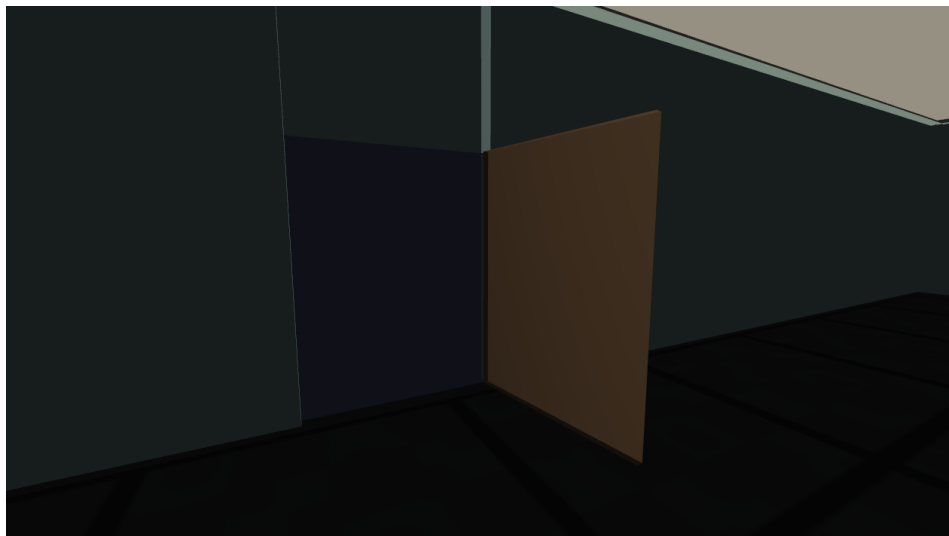


Figure 3: Video Demonstration of the Classroom Door Animation

[Click Here to Watch the Door Animation Video](#)

The video demonstrates the complete opening and closing movement of the classroom door.

The door is not a static object; its orientation changes continuously through rotational animation around the vertical axis. Therefore, the door animation provides visual evidence of the moving-object requirement of the project.

## 9.3 Lighting and Shadow Result

The classroom uses multiple lighting techniques to illuminate the environment. The lighting system can also be controlled interactively.

Figure 4 compares the classroom when the lighting system is turned on and when it is turned off.



(a) Classroom Lights ON



(b) Classroom Lights OFF

Figure 4: Comparison of the Classroom Lighting in ON and OFF States

The two images demonstrate the interactive lighting system.

When the classroom lights are enabled, the interior receives additional illumination. When the lights are disabled, the difference in scene illumination becomes clearly visible.

The project also uses different lighting techniques, including classroom point lights, projector spotlight, and directional sunlight.

Shadow effects provide additional depth to the three-dimensional environment.

#### 9.4 Projector Result

The classroom projector is an interactive component of the final environment.

When the projector is active, it directs light toward the projection screen. In presentation mode, the screen displays the **WELCOME** content as shown in Figure 5.



Figure 5: Projector Presentation Mode Displaying WELCOME

The projector provides a visible directional projection effect. The presentation screen gives visual feedback when presentation mode is active.

#### 9.5 Computer Workstations and Classroom Switches

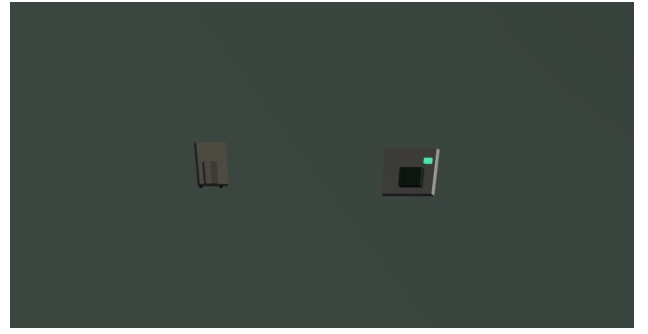
The classroom contains computer workstations and interactive wall switches.

These components improve both the visual completeness and interactivity of the classroom environment.

Figure 6 shows the computer workstation and the classroom wall switch separately.



(a) Computer Workstation



(b) Interactive Wall Switch

Figure 6: Computer Workstation and Interactive Classroom Switch

The computer workstation includes the monitor, computer tower, keyboard, mouse, and other workstation components.

The computer system also supports interactive power states.

The wall switch provides interactive control of the classroom lighting. Its visual state changes according to the current lighting condition, providing clear feedback to the user.

## 9.6 Requirement Result Summary

The final results demonstrate that the major academic requirements of the project have been implemented successfully.

| No. | Requirement                 | Status | Result Evidence                                                                                                                   |
|-----|-----------------------------|--------|-----------------------------------------------------------------------------------------------------------------------------------|
| 1   | 3D Transformation           | Yes    | Classroom objects use translation, rotation, and scaling.                                                                         |
| 2   | Viewing Transformation      | Yes    | The movable camera allows the classroom to be viewed from different positions and directions.                                     |
| 3   | Moving Object               | Yes    | The classroom door opens and closes using rotational animation. A video demonstration is provided in Section 9.2.                 |
| 4   | At Least Two Kinds of Light | Yes    | The scene uses point lights, projector spotlight, and directional sunlight. The lighting ON and OFF states are shown in Figure 4. |
| 5   | Different Colors            | Yes    | Different colors and materials distinguish the classroom objects and surfaces.                                                    |

Table 1: Final Project Requirement Verification

Overall, the final results show the required computer graphics concepts in a complete interactive environment.

The classroom can be viewed from different positions, the door provides visible animation, multiple lighting techniques illuminate the scene, and different colors and materials distinguish the classroom objects.

The additional computer workstation, wall switch, shadow, and projector features further improve the interactivity and visual completeness of the final 3D classroom.

## 10. Discussion

The project demonstrates how several fundamental computer graphics concepts can work together inside a single interactive environment.

Model transformations control the position, orientation, and dimensions of classroom objects.

Viewing transformation controls how the complete environment is observed.

The animated door demonstrates dynamic rotation transformation, while the camera demonstrates real-time viewing transformation.

The lighting system goes beyond basic illumination by combining point lights, projector spotlight, and directional sunlight.

Shadow rendering further improves the perception of depth.

The use of different colors, textures, and materials helps distinguish the floor, ceiling, desks, computers, projector, switches, and other classroom objects.

Interactive computers, wall switches, audio feedback, and projector controls demonstrate how graphical rendering can be combined with user interaction.

Overall, the final classroom demonstrates both the theoretical requirements of the course and their practical integration into a complete real-time application.

## 11. Limitations

Although the project provides an interactive and visually detailed classroom, some limitations remain.

The environment uses simplified collision and interaction logic rather than a complete physics engine.

Some classroom objects use procedural or simplified geometry instead of highly detailed professional 3D assets.

The lighting and shadow system is designed for a classroom-scale real-time application and does not provide physically accurate global illumination.

The number of fully animated objects is also limited.

Future improvements could include:

- More detailed professional 3D assets.
- Improved collision and physics simulation.
- Additional animated classroom objects.
- More advanced environment lighting.
- Additional projector presentation content.
- More detailed computer interactions.
- Improved material and surface realism.
- Additional sound and user interaction effects.

## 12. Conclusion

The Interactive 3D Classroom project successfully demonstrates the fundamental concepts required for the computer graphics project.

Translation, rotation, and scaling are used for 3D transformation, while the camera system provides viewing transformation.

The animated classroom door satisfies the moving-object requirement.

The project contains multiple lighting techniques, including point lights, projector spotlight, and directional sunlight. Different colors and material properties are used throughout the classroom.

In addition to the basic requirements, the final system includes directional sunlight, shadow rendering, textured materials, interactive computers, realistic wall switches, projector interaction, a projected **WELCOME** presentation, audio feedback, and multiple user controls.

By combining C++, OpenGL, GLM, GLFW, GLAD, GLSL, Assimp, miniaudio, and CMake, the project provides practical experience in designing, rendering, and controlling a complete interactive three-dimensional environment.

## 13. References

- [1] LearnOpenGL, <https://learnopengl.com/>
- [2] OpenGL Documentation, <https://www.opengl.org/>
- [3] GLFW Documentation, <https://www.glfw.org/documentation.html>
- [4] OpenGL Mathematics (GLM), <https://github.com/g-truc/glm>
- [5] Khronos OpenGL Reference, <https://registry.khronos.org/OpenGL/>
- [6] Assimp Documentation, <https://assimp.org/>
- [7] CMake Documentation, <https://cmake.org/documentation/>

## 14. Appendix – Source Code

The complete source code of the Interactive 3D Classroom project is publicly available on GitHub for review, reference, and further evaluation..

### Source Code Repository:

<https://github.com/shahriars41f/Interactive-3D-Classroom>